Investigation Total: ___%   Take-home: ___/40   = ___%

Year 8 Mathematics 2015

Investigation 2 – Take Home Transformations

Name:______________________________ **Due:** 27/08/15

Calculators are allowed.

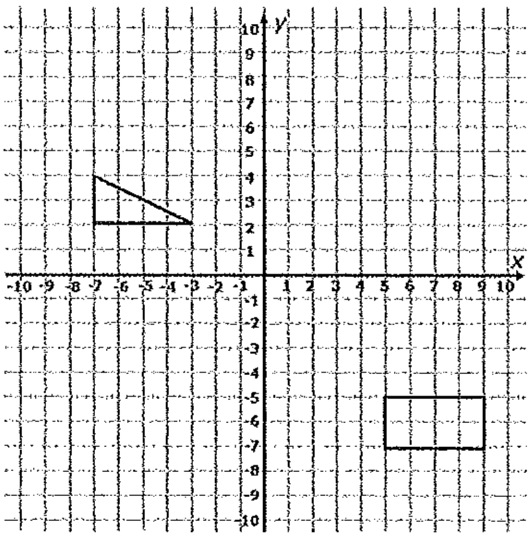
To complete this investigation, you may need to research and/or apply knowledge learned previously in class. It is up to you to decide whether to search for answers online from **reputable** sources, in your textbooks, using other sources, or by applying previous knowledge.

Part 1 Simple translations (8 marks)

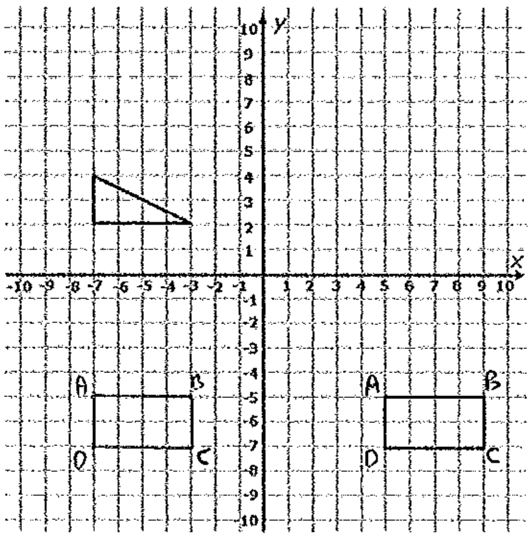
Translations are where objects move, change shape, turn, or flip according to a rule.

When performing transformations, there is some sort of shape which we start with, and we should label each corner of that shape with a letter. Consider the following shape ABCD.
Once we transform a shape, say 12 units left, we then re-label the shape - consider A'B'C'D'.

Typical shape ABCD



Translation of 12 units left to A'B'C'D'
(Note: all points (e.g. A to A') moved 12 units left.)



We always show points in the form (x,y)
So A is the point (5,-5)
B is (9,-5),
C is (9,-7),
D is (5,-7)

To describe the transformation we performed on the previous page (12 units left), we could represent it as [-12,0], where:

The left hand number shows the amount of units moved right if it is positive (and left if negative)
The right hand number shows the amount of units moved up if it is positive (and down if negative)

So, a transformation of 5 units right and 2 units down would be represented as [5,-2]

This type of transformation we have performed is known as a translation

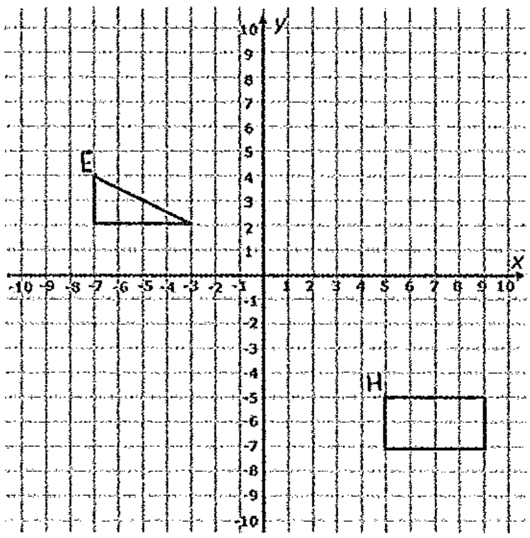
Question 1 **(1,1,1,5 = 8 marks)**

a) Define what a translation means in the context of maths (geometry)

b) Describe a transformation of 4 units left and 3 units up using notation [#,#] shown above

c) Describe a transformation of 2 units down and 3 units right using notation [#,#]

d) On the following diagram, label the triangle EFG and label the rectangle HIJK. Translate the triangle 2 units left and 5 units up. Translate the rectangle [1,10]. Label your new shapes E'F'G' and follow the same process for the rectangle.



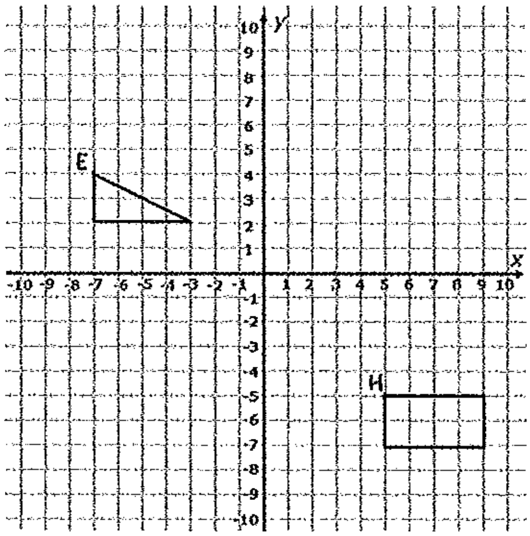
Note: the 'new shapes' (e.g. E'F'G') are the referred to as the images of the original shapes

Part 2 Reflections (15 marks)

Question 2 **(1,2,2,3 = 8 marks)**

a) Define what a reflection means in the context of maths (geometry)

b) Reflect both of the shapes **in the x-axis** (by reflecting each point and then joining them).



c) Complete the following table to compare coordinates of initial points with images (label!)

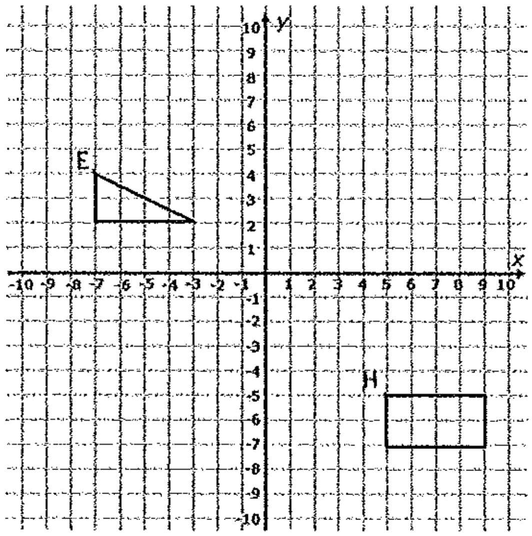
Triangle			
Original Letter	Coordinates	Image	Coordinates
E	(-7,4)	E'	(-7,-4)

Square			
Original Letter	Coordinates	Image	Coordinates
H	(5,-5)	H'	(5,5)

d) Describe with words what has happened to the x and y-coordinates for each point (x,y) in a reflection in the x axis. See if you can come up with a rule for an x-axis reflection as [#,#]

Question 3 **(2,2,3 = 7 marks)**

a) Reflect both of the shapes **in the y-axis** (by reflecting each point and then joining them).



b) Complete the following table to compare coordinates of initial points with images

Triangle			
Original Letter	Coordinates	Image	Coordinates

Square			
Original Letter	Coordinates	Image	Coordinates

c) Describe with words what has happened to the x and y-coordinates for each point (x,y) in a reflection in the y-axis. See if you can come up with a rule for an y-axis reflection as [#,#]

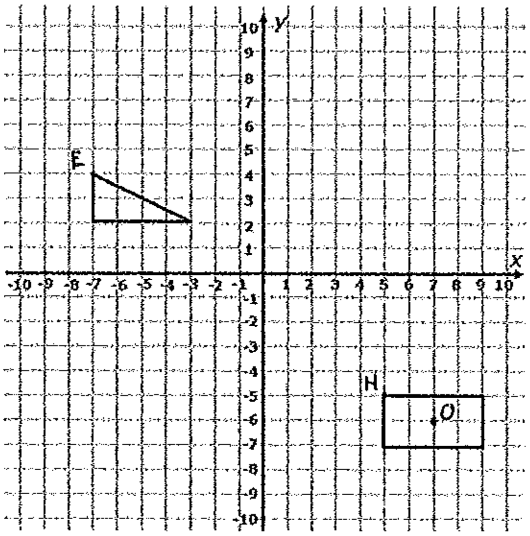
Part 3 Rotations **(8 marks)**

Question 4 **(1,2,2,1,1,1 = 8 marks)**

a) Define what a rotation means in the context of maths (geometry)

b) Rotate the triangle about the origin (i.e. about the point (0,0)) by 90° clockwise

c) Rotate the rectangle about the point O by 90° anticlockwise



d) Draw a shape by plotting the points L (1,1), M (2,3) and N (3,1). Join these to form the shape LMN

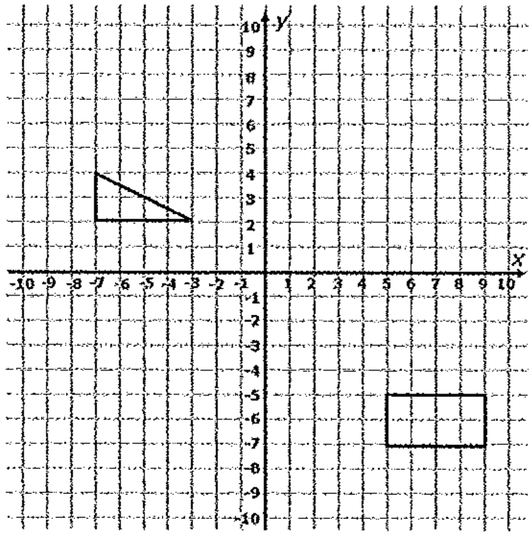
e) Reflect LMN in the x-axis to form L'M'N' and then reflect L'M'N' in the y-axis to form L"M"N"

f) Could the transformation from LMN to L"M"N" have been performed by any other single transformation? If so, specify exactly what would have needed to be done to get there.

Part 4 Combined transformations (9 marks)

Question 5 **(2,2,2,2,1 = 9 marks)**

a) Plot O (0,5), P (1,8), Q (3,5) and R(4,8). Join the points to form OPQR.



b) Perform the translation [4,-4] and label the image.

c) Reflect O'P'Q'R' in the x-axis. Label the new image.

d) Rotate O"P"Q"R" about the origin (0,0) 90 degrees clockwise. Label the new image.

e) What type of shape is O'''P'''Q'''R'''? Has it OPQR changed shape through this process?